

Titration report



Method

Method ID	CDC_M003	Release state	Released
Name	Titration with HCl - Liquid	Modified by	PredefinedUser
Modified on	2/25/2026 8:38:44 AM		

Summary

Samples

Sample ID	Name	Result	State
1	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.014121 mL
		Concentration R2	0.10028 mol/L
2	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.013111 mL
		Concentration R2	0.10026 mol/L
3	0.1 M TRIS - 5 mL in 60 mL H2O	Sample size	5 mL
		Consumption R1	5.010497 mL
		Concentration R2	0.10021 mol/L

Statistics

R1

Mean	5.012576 mL	Minimum	5.010497 mL
Standard deviation	0.001870 mL	Rel. standard deviation	0.037 %
Total samples	3	Samples excluded	0

R2

Mean	0.10025 mol/L	Minimum	0.10021 mol/L
Standard deviation	0.00004 mol/L	Rel. standard deviation	0.036 %
Total samples	3	Samples excluded	0

Task

Task internal ID	T165235	Task name	New electrode - UNUSED
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Main Results

Sample ID1	Sample ID2	Result Title	Name	Result	State
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.014121 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Concentration	R2	0.10028 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O		Consumption	R1	5.013111 mL	OK

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0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10026 mol/L	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Consumption	R1	5.010497 mL	OK
0.1 M TRIS - 5 mL in 60 mL H2O	Concentration	R2	0.10021 mol/L	OK

Raw Results

Sample scope: Scope 1/1, Sample 1/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 8:42:07 AM
Overall state	OK
Sample start time	2/25/2026 8:42:07 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 8:42:11 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 8:42:11 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:13 min

Stir (Stir1)

Creation time	2/25/2026 8:42:38 AM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 8:44:39 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	103.7 mV
EHNv1	-72.4 mV
Q1	0.501412 mmol
VEQ1	5.014121 mL
EST	-176.5 mV
VEX	1.371379 mL
VEND	6.3855 mL
t	02:13 min
QEND	0.63855 mmol
QEX	0.137138 mmol
nEQ	1

Consumption (R1)

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Creation time	2/25/2026 8:46:57 AM
Overall state	OK
Result	5.014121 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	2/25/2026 8:46:57 AM
Overall state	OK
Result	0.10028 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 2/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 8:47:34 AM
Overall state	OK
Sample start time	2/25/2026 8:47:34 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 8:47:35 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 8:47:36 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:13 min

Stir (Stir1)

Creation time	2/25/2026 8:48:03 AM
Overall state	OK

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 8:50:04 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	102.5 mV
EHNv1	-72.2 mV
Q1	0.501311 mmol
VEQ1	5.013111 mL

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EST	-176.7 mV
VEX	1.283389 mL
VEND	6.2965 mL
t	02:09 min
QEND	0.62965 mmol
QEX	0.128339 mmol
nEQ	1

Consumption (R1)

Creation time	2/25/2026 8:52:17 AM
Overall state	OK
Result	5.013111 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

Creation time	2/25/2026 8:52:18 AM
Overall state	OK
Result	0.10026 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Sample scope: Scope 1/1, Sample 3/3 (0.1 M TRIS - 5 mL in 60 mL H2O)

Sample (S1)

Creation time	2/25/2026 8:52:45 AM
Overall state	OK
Sample start time	2/25/2026 8:52:45 AM
Sample size	5 mL
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/25/2026 8:52:46 AM
Overall state	OK

Dispense (normal) [1] (Dispensenormal1)

Creation time	2/25/2026 8:52:47 AM
Overall state	OK
VENDDi	5 mL
QENDDi	0.5 mmol
tDi	00:12 min

Stir (Stir1)

Creation time	2/25/2026 8:53:14 AM
Overall state	OK

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Titration (EQP) [1] (TitrationEQP1)

Creation time	2/25/2026 8:55:14 AM
Overall state	OK
Terminated at	Number of EQPs
EEQ1	98.9 mV
EHN1	-72.8 mV
Q1	0.501050 mmol
VEQ1	5.010497 mL
EST	-176.3 mV
VEX	1.203003 mL
VEND	6.2135 mL
t	02:18 min
QEND	0.62135 mmol
QEX	0.120300 mmol
nEQ	1

Consumption (R1)

Creation time	2/25/2026 8:57:38 AM
Overall state	OK
Result	5.010497 mL
Decimal places	6
Substance	None
M	1 g/mol
z	1

Concentration (R2)

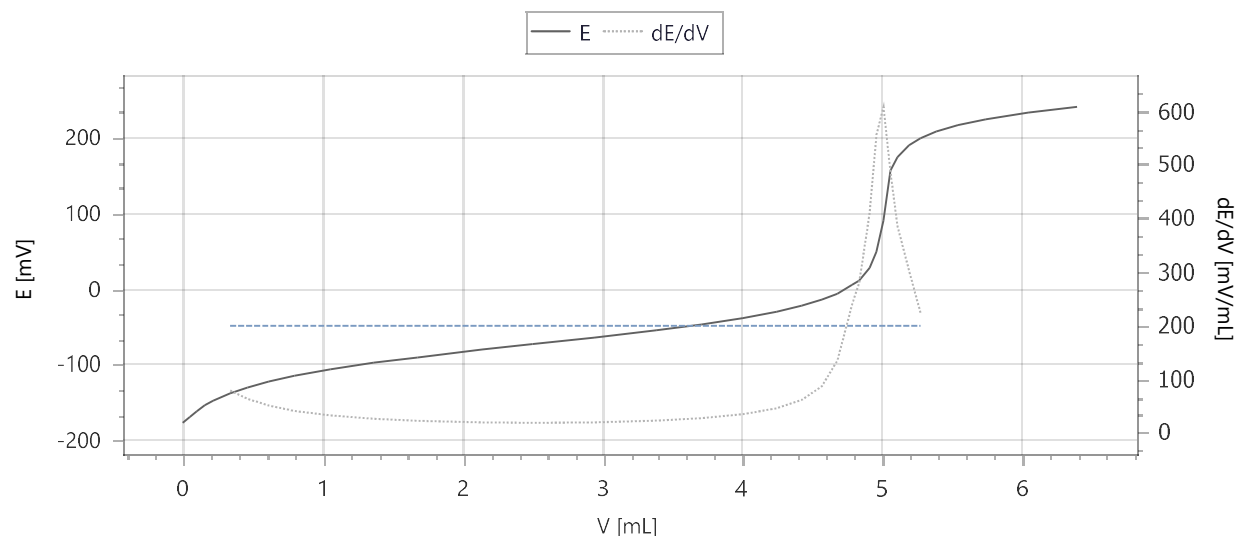
Creation time	2/25/2026 8:57:38 AM
Overall state	OK
Result	0.10021 mol/L
Decimal places	5
Substance	None
M	1 g/mol
z	1

Titration (EQP) [1] Scope 1/1, Sample 1/3

E - V & dE/dV - V Curve Scope 1/1, Sample 1/3

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Measured Values Scope 1/1, Sample 1/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.5		25.0			
3	0.0500	-168.7	0.0500	25.0	7.8		
6	0.1000	-161.2	0.0500	25.0	7.5		
9	0.1570	-153.2	0.0570	25.0	8.0		
12	0.2170	-147.4	0.0600	25.0	5.8		
15	0.3390	-137.8	0.1220	25.0	9.6	79.32	
18	0.4610	-129.9	0.1220	25.0	7.9	64.08	
21	0.6140	-121.8	0.1530	25.0	8.1	51.49	
24	0.7970	-114.3	0.1830	25.0	7.5	41.38	
28	1.0535	-106.1	0.2565	25.0	8.2	33.13	
30	1.3655	-97.1	0.3120	25.0	9.0	26.81	
34	1.6785	-90.1	0.3130	25.0	7.0	23.00	
37	2.1380	-79.9	0.4595	25.0	10.2	20.11	
40	2.4985	-72.6	0.3605	25.0	7.3	19.31	
43	2.9360	-64.2	0.4375	25.0	8.4	20.04	
46	3.3815	-54.5	0.4455	25.0	9.7	23.37	
49	3.7025	-46.8	0.3210	25.0	7.7	27.91	
52	4.0000	-38.4	0.2975	25.0	8.4	35.43	
55	4.2425	-29.7	0.2425	25.0	8.7	46.32	
58	4.4205	-21.8	0.1780	25.0	7.9	61.89	
61	4.5640	-13.7	0.1435	25.0	8.1	87.36	
64	4.6750	-6.3	0.1110	25.0	7.4	135.54	
68	4.7795	5.7	0.1045	25.0	12.0	239.03	
71	4.8295	11.3	0.0500	25.0	5.6	278.32	
76	4.9040	28.2	0.0745	25.0	16.9	407.88	
79	4.9540	49.3	0.0500	25.0	21.1	555.56	
83	5.0040	90.3	0.0500	25.0	41.0	607.05	
	5.014121	103.7				607.63	EQP
89	5.0540	156.3	0.0500	25.0	66.0	485.11	
93	5.1040	174.2	0.0500	25.0	17.9	385.21	
97	5.1865	189.7	0.0825	25.0	15.5	302.43	

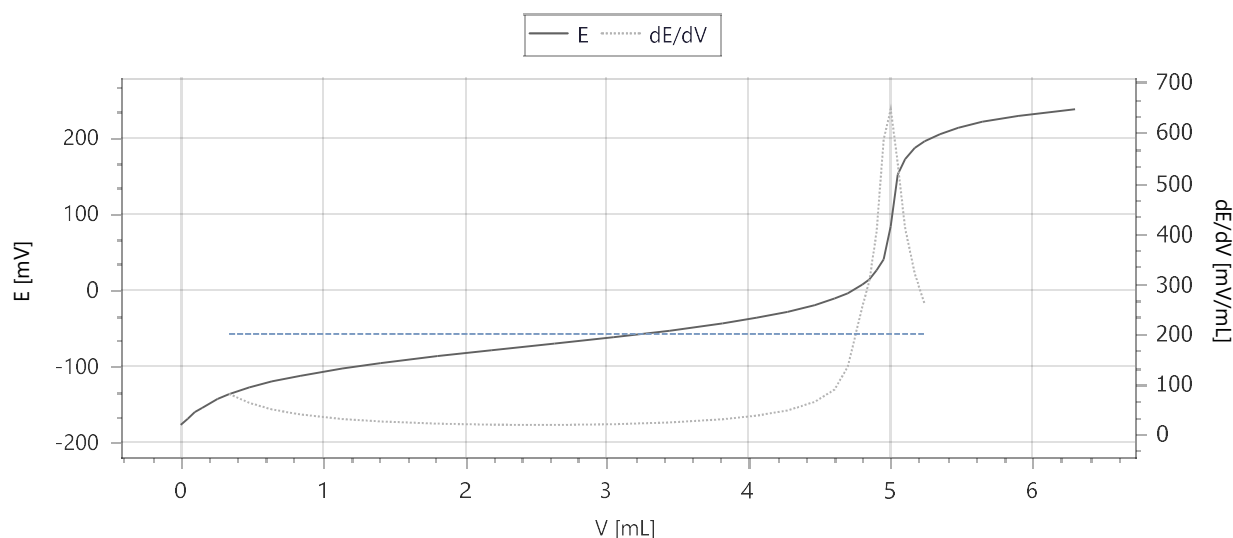
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100	5.2680	199.1	0.0815	25.0	9.4	224.01
103	5.3795	207.8	0.1115	25.0	8.7	
106	5.5355	216.4	0.1560	25.0	8.6	
109	5.7350	224.1	0.1995	25.0	7.7	
113	6.0375	233.1	0.3025	25.0	9.0	
116	6.3855	240.4	0.3480	25.0	7.3	

Titration (EQP) [1] Scope 1/1, Sample 2/3

E - V & dE/dV - V Curve Scope 1/1, Sample 2/3



Measured Values Scope 1/1, Sample 2/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.7		25.0			
3	0.0500	-168.6	0.0500	25.0	8.1		
6	0.1000	-159.8	0.0500	25.0	8.8		
9	0.1500	-154.2	0.0500	25.0	5.6		
12	0.2590	-142.7	0.1090	25.0	11.5		
15	0.3405	-136.4	0.0815	25.0	6.3	80.72	
18	0.4815	-127.6	0.1410	25.0	8.8	63.04	
21	0.6430	-119.5	0.1615	25.0	8.1	49.73	
24	0.8395	-112.3	0.1965	25.0	7.2	40.05	
28	1.1360	-102.6	0.2965	25.0	9.7	30.75	
30	1.4095	-95.5	0.2735	25.0	7.1	25.98	
34	1.8075	-86.2	0.3980	25.0	9.3	21.62	
37	2.1850	-78.5	0.3775	25.0	7.7	19.56	
40	2.6315	-69.7	0.4465	25.0	8.8	19.01	
43	3.0490	-61.3	0.4175	25.0	8.4	20.51	
46	3.4430	-53.1	0.3940	25.0	8.2	23.99	
49	3.8145	-43.4	0.3715	25.0	9.7	30.30	
52	4.0575	-35.9	0.2430	25.0	7.5	37.60	
55	4.2720	-28.1	0.2145	25.0	7.8	47.77	
58	4.4645	-19.4	0.1925	25.0	8.7	64.91	
61	4.6035	-10.7	0.1390	25.0	8.7	89.37	

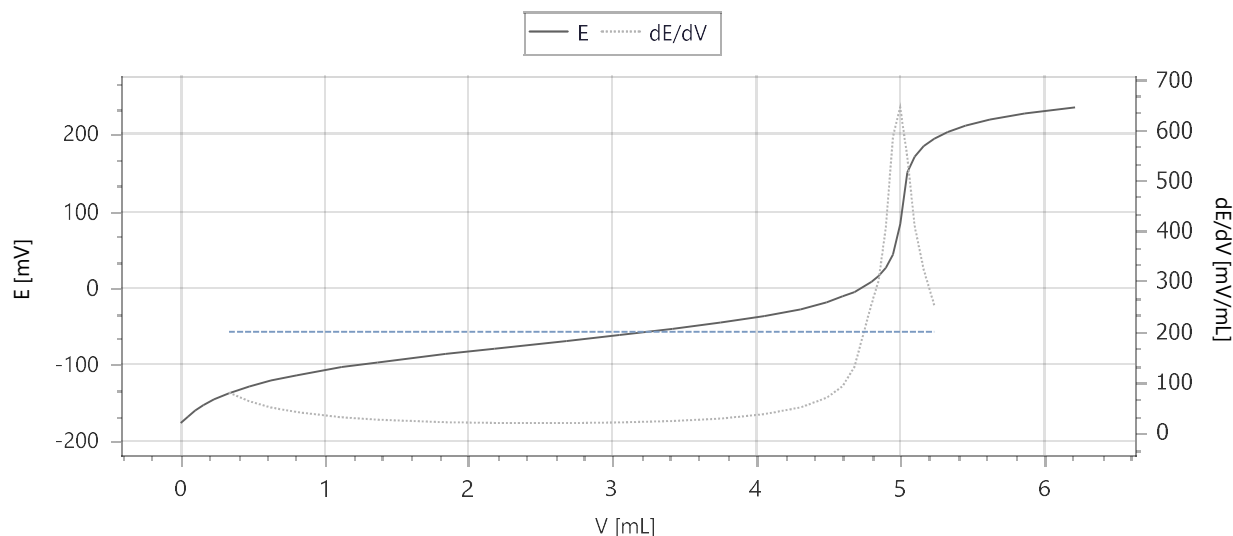
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LabX

64	4.6970	-4.1	0.0935	25.0	6.6	134.80	EQP
68	4.7985	7.6	0.1015	25.0	11.7	253.29	
70	4.8485	14.3	0.0500	25.0	6.7	305.07	
74	4.8995	26.2	0.0510	25.0	11.9	402.24	
77	4.9495	40.5	0.0500	25.0	14.3	586.04	
80	4.9995	83.9	0.0500	25.0	43.4	646.47	
	5.013111	102.5				648.15	
87	5.0495	152.1	0.0500	25.0	68.2	536.26	
90	5.0995	172.0	0.0500	25.0	19.9	410.93	
94	5.1685	186.9	0.0690	25.0	14.9	319.82	
97	5.2375	195.6	0.0690	25.0	8.7	258.77	
100	5.3450	205.1	0.1075	25.0	9.5		
103	5.4755	213.5	0.1305	25.0	8.4		
106	5.6445	221.2	0.1690	25.0	7.7		
109	5.8950	228.7	0.2505	25.0	7.5		
112	6.2965	237.9	0.4015	25.0	9.2		

Titration (EQP) [1] Scope 1/1, Sample 3/3

E - V & dE/dV - V Curve Scope 1/1, Sample 3/3



Measured Values Scope 1/1, Sample 3/3

Time [s]	V [mL]	Meas. value [mV]	dV [mL]	Temp. [°C]	dE [mV]	dE/dV [mV/mL]	EQP
0	0.0000	-176.3		25.0			
3	0.0500	-167.8	0.0500	25.0	8.5		
6	0.1000	-160.1	0.0500	25.0	7.7		
9	0.1575	-152.6	0.0575	25.0	7.5		
12	0.2310	-145.3	0.0735	25.0	7.3		
15	0.3350	-137.4	0.1040	25.0	7.9	79.92	
18	0.4760	-128.5	0.1410	25.0	8.9	62.73	
21	0.6260	-120.6	0.1500	25.0	7.9	50.45	
24	0.8070	-113.9	0.1810	25.0	6.7	41.31	
28	1.1190	-103.1	0.3120	25.0	10.8	31.01	
30	1.3670	-97.2	0.2480	25.0	5.9	26.35	

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34	1.8420	-85.8	0.4750	25.0	11.4	21.26	EQP
37	2.1790	-79.4	0.3370	25.0	6.4	19.96	
40	2.6790	-69.3	0.5000	25.0	10.1	19.57	
43	3.0485	-61.5	0.3695	25.0	7.8	20.90	
46	3.4165	-53.3	0.3680	25.0	8.2	23.70	
49	3.7510	-45.2	0.3345	25.0	8.1	28.47	
52	4.0545	-36.8	0.3035	25.0	8.4	37.24	
55	4.3085	-27.8	0.2540	25.0	9.0	50.71	
58	4.4875	-18.8	0.1790	25.0	9.0	69.57	
61	4.5990	-11.0	0.1115	25.0	7.8	92.58	
64	4.6810	-5.4	0.0820	25.0	5.6	130.71	
69	4.7990	8.1	0.1180	25.0	13.5	254.19	
72	4.8490	15.3	0.0500	25.0	7.2	301.25	
78	4.8990	26.4	0.0500	25.0	11.1	405.66	
84	4.9490	43.6	0.0500	25.0	17.2	585.73	
90	4.9990	83.4	0.0500	25.0	39.8	643.75	
	5.010497	98.9				644.04	
97	5.0490	150.8	0.0500	25.0	67.4	543.99	
100	5.0990	171.4	0.0500	25.0	20.6	409.67	
104	5.1625	184.9	0.0635	25.0	13.5	322.55	
106	5.2355	194.7	0.0730	25.0	9.8	252.49	
110	5.3295	203.5	0.0940	25.0	8.8		
113	5.4535	211.7	0.1240	25.0	8.2		
116	5.6225	219.6	0.1690	25.0	7.9		
119	5.8650	227.6	0.2425	25.0	8.0		
122	6.2135	235.8	0.3485	25.0	8.2		

Sample Series

Name	Internal ID
Label 1	Label 2
Label 3	Label 4
Label 5	Label 6

Comment

Sample

Sample ID 1
Sample ID 2
Sample ID 3

Sample Parameters

Type	Name	Qualified name	Value
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Task Parameter

Sample ID	Parameter name	Qualified name	Type	Value	Time
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 08:42:05

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0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	New electrode - UNUSED	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	New electrode - UNUSED	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Volume [mL]	S1	Decimal	5	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Density [g/mL]	S1	Decimal	1.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Correction factor	S1	Decimal	1.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Temperature [°C]	S1	Decimal	25.0	25.02.2026 08:42:05
0.1 M TRIS - 5 mL in 60 mL H2O	Comment	S1	String	New electrode - UNUSED	25.02.2026 08:42:05

Used Instruments

Name	T9_MSG_SA_002	Serial number	B551866924
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Responsible User

Time	Login name	Full name
25.02.2026 08:42:05	PredefinedUser	Administrator